INTAKE SHUTTER VALVE [SKYACTIV-D 1.5]

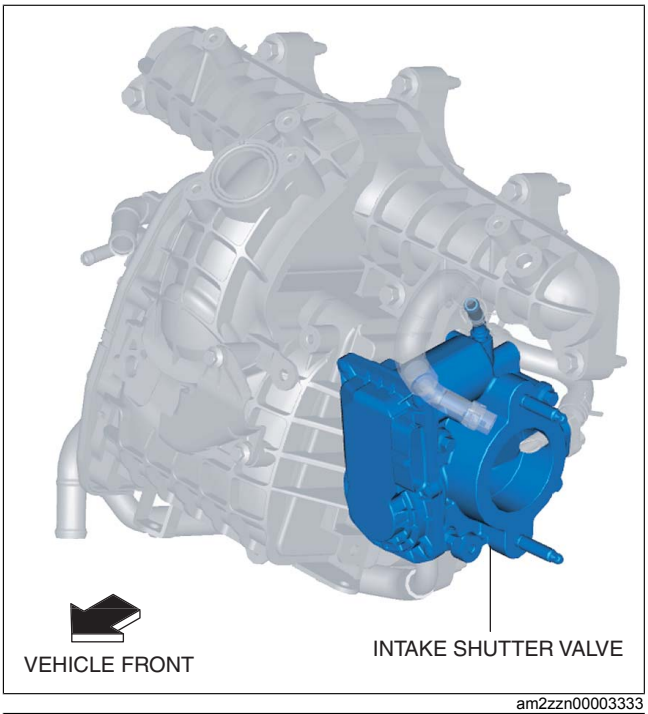
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Purpose, Function

- The intake shutter valve facilitates recirculation induction of the exhaust gas by adjusting the intake air volume and channeling the airflow to the intake manifold side.
- The intake air amount to the engine is reduced to increase the exhaust gas temperature easily while the DPF is carrying out combustion. Due to this, the particulate matter (PM) collected in the DPF is burned and eliminated.
- The valve is closed to cut the intake air when the engine switch is off to reduce the shock when the engine is turned off.

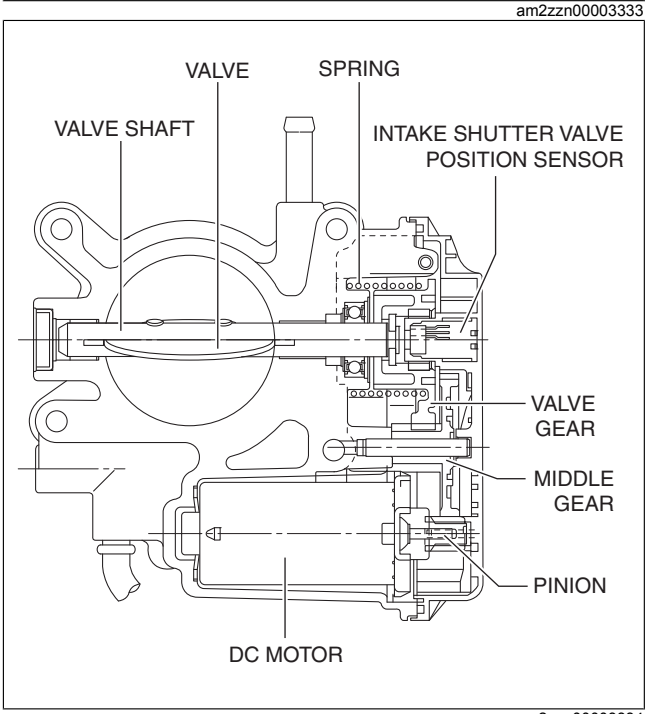
Construction

- The intake shutter valve is located on intake manifold.



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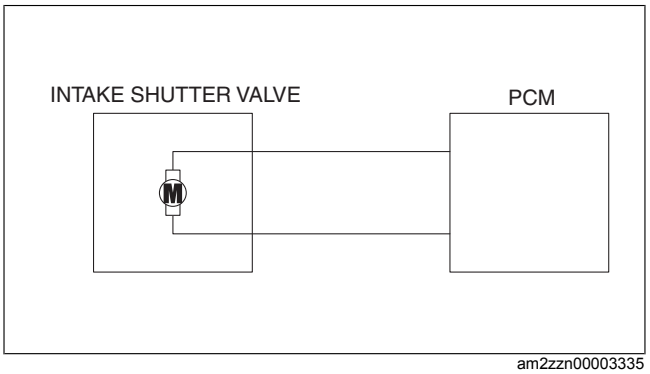
- The intake shutter valve consists of the shutter valve, valve shaft, valve, spring, valve gear, mid gear, DC motor, and pinion.
- The intake shutter valve has an integrated intake shutter valve position sensor. For details on the intake shutter valve position sensor, refer to the CONTROL SYSTEM. (See INTAKE SHUTTER VALVE POSITION SENSOR [SKYACTIV-D 1.5].)



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Operation

- The intake shutter valve is opened to the valve opening angle according to the energization current flowing to the DC motor from the PCM.
- The intake shutter valve performs coordinated control with the LP-EGR control valve, HP-EGR control valve, and exhaust shutter valve. For details on the exhaust shutter valve, refer to the CONTROL SYSTEM. (See INTAKE SHUTTER VALVE CONTROL [SKYACTIV-D 1.5].)
- If the shutter valve has a malfunction and the DC motor control is disabled, the shutter valve is opened to the fully open position by the return spring. As a result, the air amount necessary to keep the vehicle moving is assured.



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Fail-safe

DTC No.	Fail-safe function
P0122:00	• Limits the engine torque. • Inhibits the EGR control. • Fully open the intake shutter valve. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P0123:00	
P1589:00	
P2101:00	
P2118:00	
P0401:00	• Inhibits engine-stop by operating the i-stop function.
P0402:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
P2104:00	• Prevents a combustion pressure increase by closing the intake shutter valve.